

Analysis plots for ORJointUpdate (OR_joint_update2.Rmd)

Claude

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This note explains the three diagnostic plots produced by `OR_joint_update_analysis.Rmd` for the joint disturbance/process-error filter in `OR_joint_update2.Rmd`, and gives the formulas for every quantity they show:

1. mean normalized innovation squared (NIS) per site;
2. disturbance detection vs. absolute disturbance size;
3. mean innovation variance split into forecast spread, process error and observation error.

Section 5 describes the experiment grid and the 5-site test run in `experiments/testRun`, and shows its plots.

1 The filter's forecast predictive

The state is $x_t = (C_v, C_s, C_R)^\top$ (leaf, soil, wood). Only wood is observed,

$$y_t \mid x_t \sim \mathcal{N}(Hx_t, R_t), \quad H = (0, 0, 1), \quad R_t = m_R y_t + b_R,$$

with $(m_R, b_R) = \text{Rparams}$ ($m_R = 0$ for the constant- R experiments).

Process precision grid. The wood process precision τ is unknown. The filter keeps a log-spaced grid $\tau_1 < \dots < \tau_G$ ($G = 200$) with posterior weights $\pi_t(g) = p(\tau_g \mid y_{1:t})$, starting from the prior cell masses $\pi_0(g)$ and updated by

$$\pi_t(g) \propto \pi_{t-1}(g) Z_t(g), \quad Z_t(g) = \sum_{j=1}^2 \sum_{k=0}^D w_{t-1}(g, j) \beta_{k,t} \mathcal{N}(y_t; \mu_{gjk}^F, \Sigma_{gjk,33}^F + R_t).$$

In code $\log \pi_t$ is `qs`, $Z_t(g)$ is `exp(logz[g])`.

Forecast components. For each grid point g the analysis is projected onto two Gaussians $j \in \{1, 2\}$ with weights $w_{t-1}(g, j)$ (`prevW`). Pushing an importance-weighted VSEM ensemble through one year and applying disturbance label $k \in \{0, \dots, D\}$ ($k = 0$ = undisturbed, $k \geq 1$ = disturbance sub-class) gives the forecast moments $\mu_{gjk}^F, \Sigma_{gjk}^F$ (`muF`, `sigF`). Label k has prior $\beta_{k,t} = \text{dist\$p}[k+1, t]$. Write

$$C_{gjk} = \text{weighted ensemble variance of the wood pool for } (g, j, k).$$

Process error is added only to the undisturbed label, with wood variance $1/\tau_g$:

$$\Sigma_{gjk,33}^F = C_{gjk} + \mathbf{1}\{k = 0\} \tau_g^{-1}.$$

Predictive of y_t (integrating over process error). Before y_t is assimilated, the one-step-ahead predictive is the full mixture

$$p(y_t | y_{1:t-1}) = \sum_{g=1}^G \sum_{j=1}^2 \sum_{k=0}^D \omega_{gjk} \mathcal{N}(y_t; \mu_{gjk,3}^F, \Sigma_{gjk,33}^F + R_t), \quad \omega_{gjk} = \pi_{t-1}(g) w_{t-1}(g, j) \beta_{k,t}. \quad (1)$$

The sum over g uses the previous τ posterior π_{t-1} , so the process error is integrated out instead of being fixed at one value of τ ; the sum over k uses the prior $\beta_{k,t}$, because the label for year t is not known before y_t is seen. The weights sum to one ($\sum_j w = 1$, $\sum_k \beta = 1$; the code renormalises anyway).

The predictive mean and variance follow from the laws of total expectation and variance:

$$m_t = \sum_{gjk} \omega_{gjk} \mu_{gjk,3}^F, \quad (2)$$

$$S_t = \sum_{gjk} \omega_{gjk} \left[\Sigma_{gjk,33}^F + (\mu_{gjk,3}^F - m_t)^2 \right] + R_t. \quad (3)$$

2 Plot 1: mean normalized innovation squared

Formula. The innovation and the NIS at observed step t are

$$\nu_t = y_t - m_t, \quad \text{NIS}_t = \frac{\nu_t^2}{S_t},$$

and the plotted value for site s is the time average over its N observed steps,

$$\overline{\text{NIS}}_s = \frac{1}{N} \sum_{t=1}^N \text{NIS}_t.$$

Stored per step in `hist$innov` and `hist$nis`.

What changed vs. `OR_joint_update.Rmd`. The older filter computed NIS against one Gaussian: it kept only the label with the largest *posterior* weight and collapsed that label's forecast across Gauss-Laguerre nodes. Here m_t and S_t are the moments of the whole forecast predictive (1). The new version differs in three ways: (i) the integral over τ uses the grid with the *previous* posterior π_{t-1} , so it does not use y_t ; (ii) the disturbance labels are weighted by their *prior* $\beta_{k,t}$, again without using y_t ; (iii) the two projected components j are included. As a result, NIS_t is a true one-step-ahead (out-of-sample) check.

How to read it. Because m_t and S_t are the exact mean and variance of the predictive, a calibrated filter gives $\mathbb{E}[\text{NIS}_t] = 1$ *whatever the shape* of the predictive. Values above 1 mean the forecast is overconfident or biased; values below 1 mean the predictive is too wide. The dashed line marks 1. The grey band is

$$\left[\chi_{N,0.025}^2/N, \chi_{N,0.975}^2/N \right],$$

the 95% range of $\overline{\text{NIS}}$ when the predictive is Gaussian and the steps are independent. Here the predictive is a mixture: a scale mixture over τ plus the disturbance components. That makes NIS_t heavier-tailed than χ_1^2 . Typical values then sit below 1, with an occasional large value in a year the forecast did not expect. So treat the band as a guide, not an exact test.

Panels. Page 1 plots $\overline{\text{NIS}}_s$ (log scale) against the final-year observed biomass, one panel per experiment. Page 2 puts all experiments on one axis. Points are coloured by the 2004 disturbance-product label. Page 6 shows the numerator and denominator separately: bars are $\frac{1}{N} \sum_t \nu_t^2$ and points are $\frac{1}{N} \sum_t S_t$ for each site.

3 Plot 2: disturbance detection vs. absolute disturbance size

Detection. The posterior probability that site s was disturbed in 2004 (t_{04} = timestep 20) is

$$P_s = \Pr(\text{disturbed in } t_{04} \mid y_{1:t_{04}}) = \sum_g \pi_{t_{04}}(g) \sum_j \sum_{k \geq 1} w_{t_{04}}^A(g, j, k),$$

$$w_t^A(g, j, k) = \frac{w_{t-1}(g, j) \beta_{k,t} \mathcal{N}(y_t; \mu_{gjk,3}^F, \Sigma_{gjk,33}^F + R_t)}{Z_t(g)},$$

i.e. `hist$pDist[t04]`. (This is the filtering probability, so it uses data up to 2004 only.)

Size. The x-axis is the absolute observed change in above-ground biomass between 2003 and 2004:

$$\Delta_s = y_{s,2003} - y_{s,2004} \quad (\text{kg C m}^{-2}),$$

which is positive for a loss. The sign is kept so that gains are visible. Points are coloured by the disturbance-product label for 2004 (`dextract[, "t2004"]`: none, other, cut, fire, biotic).

Fitted curve (full runs only). When an experiment has at least 20 sites, the script fits a beta regression

$$\tilde{P}_s \sim \text{Beta}(\mu_s \phi, (1 - \mu_s) \phi), \quad \text{logit } \mu_s = a + b \Delta_s, \quad \tilde{P}_s = \frac{P_s(n-1) + 0.5}{n},$$

where $\tilde{P}$ is shrunk so that it stays inside $(0, 1)$. The plot shows $\mu(\Delta)$ with a 2.5–97.5% beta quantile band. Page 7 tabulates the size detected with probability one half, $\Delta_{50} = -a/b$. With 5 sites no fit is made, and the page shows only the points.

Reading it. With the product, the prior $\beta_{k,2004}$ is raised to the product’s user’s accuracy in years the product flags. So a flagged site can have a high P_s even when Δ_s is small. Without the product, P_s depends only on how well Δ_s stands out against S_t .

4 Plot 3: innovation variance decomposition

Equation (3) splits into three additive parts, $S_t = S_t^{fc} + S_t^q + S_t^R$:

$$\begin{aligned} S_t^q &= \sum_{g,j} \omega_{gj0} \tau_g^{-1} = \beta_{0,t} \sum_g \pi_{t-1}(g) \tau_g^{-1} = \beta_{0,t} \mathbb{E}_{\pi_{t-1}}[\tau^{-1}] && (\text{process error}) \\ S_t^{fc} &= \underbrace{\sum_{gjk} \omega_{gjk} C_{gjk}}_{\text{ensemble}} + \underbrace{\sum_{gjk} \omega_{gjk} (\mu_{gjk,3}^F - m_t)^2}_{\text{between components}} && (\text{forecast spread}) \\ S_t^R &= R_t = m_R y_t + b_R && (\text{observation error}) \end{aligned}$$

Stored per step as `hist$S.fc`, `hist$S.q`, `hist$S.R`. In code, S_t^{fc} is computed as $\sum \omega(\Sigma_{33}^F + (\mu_3^F - m_t)^2) - S_t^q$.

Process error enters only through the undisturbed label. It is the posterior-mean process variance scaled by the prior probability of no disturbance. The between-component term in S^{fc} contains the spread between disturbed and undisturbed forecasts, weighted by $\beta_{k,t}$. In years where the disturbance prior is large, for example product-flagged years, this term can be much larger than the others.

Panels. Each part is averaged over a site’s observed steps, $\bar{S}_s = \frac{1}{N} \sum_t S_t$. Page 4 shows per-site stacked bars (all sites when there are at most 30, otherwise the 30 largest). Page 5 shows, for each experiment, the across-site mean of each part (left) and each part’s share of the total (right).

5 Experiment grid and test run

The run-grid chunk of `OR_joint_update2.Rmd` runs all $2 \times 2 \times 2 = 8$ experiments:

factor	levels
R magnitude	GEDI, LandTrendr
R model	heteroskedastic (<code>het</code>), constant (<code>con</code>)
disturbance product	TRUE, FALSE

`Rparams` (m_R, b_R): GEDI `het` (0.0133, 0.0976), GEDI `con` (0, 0.885), LandTrendr `het` (0.0133, 3.013), LandTrendr `con` (0, 3.800).

The τ prior is `Gamma(0.1,  $2 \times 10^{-4}$ )` (`grid.tauPrior = "gamma"`). The test run used one randomly chosen site (seed 2004) for each 2004 product label, giving 5 sites (`grid.sites`). Runs are written to `experiments/testRun` as

```
site<idx>_distprod<TF>_<source>_<model>_tau<prior>.rds,
```

and `OR_joint_update_analysis.Rmd` (with `indir` set to `experiments/testRun`) writes `exps_summary.pdf` there. For the full run, set `grid.sites` to all sites with `grid.sites <- seq_len(nrow(dextract))`.

Test-run results (sites 11, 118, 212, 319, 402)

The five sites have 2004 labels none, other, cut, fire and biotic. The figures below are pages 1, 3, 4 and 5 of `experiments/testRun/exps_summary.pdf`. With five sites these plots show what each panel looks like and give a first sanity check; they are not a calibration result.

- **Mean NIS.** Four of the five sites are below 1 in every experiment (0.15–0.8), so the predictive is too wide at those sites. The between-component spread in S^{fc} is a large part of the reason. Site 212 (cut) is at 1.9–3.8. Its pixel was clear-cut in 1999 ($19.4 \rightarrow 1.6$ kg C m⁻²), which gives $\text{NIS} \approx 57$ in that one year. The filter then labels the drop as a disturbance ($P = 1$). Its forecasts also lag the fast regrowth from 2007 to 2009 (NIS 5–15). Every site also has a large innovation at $t = 1$, because the initial-condition ensemble is the cross-site AGB distribution.
- **Detection.** The observed 2003→2004 changes at these sites are small (−0.1 to 0.45 kg C m⁻²). Without the product, $P_{2004} \leq 0.05$ everywhere. With the product, the four flagged sites reach 0.49–0.96 and the undisturbed site stays at 0.01–0.03. So at these sites the product prior, not the data, drives detection.

- **Variance decomposition.** With GEDI-magnitude R , S is mostly forecast spread (65–88%). With LandTrendr-magnitude R , observation error is about half of S . Process error S^q is the smallest part everywhere (2–11%). It is smaller still with the product, because a raised $\beta_{k,t}$ lowers the weight $\beta_{0,t}$ on the only label that carries process error.

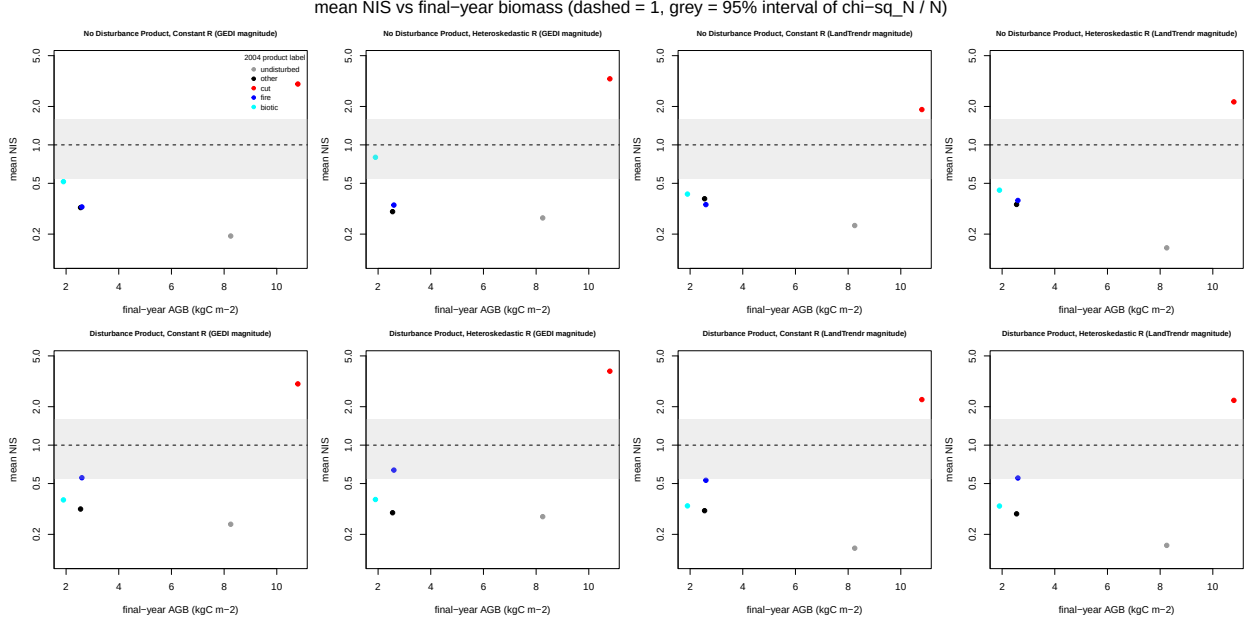


Figure 1: Mean NIS vs. final-year biomass, one panel per experiment (test run).
disturbance detection vs absolute disturbance size

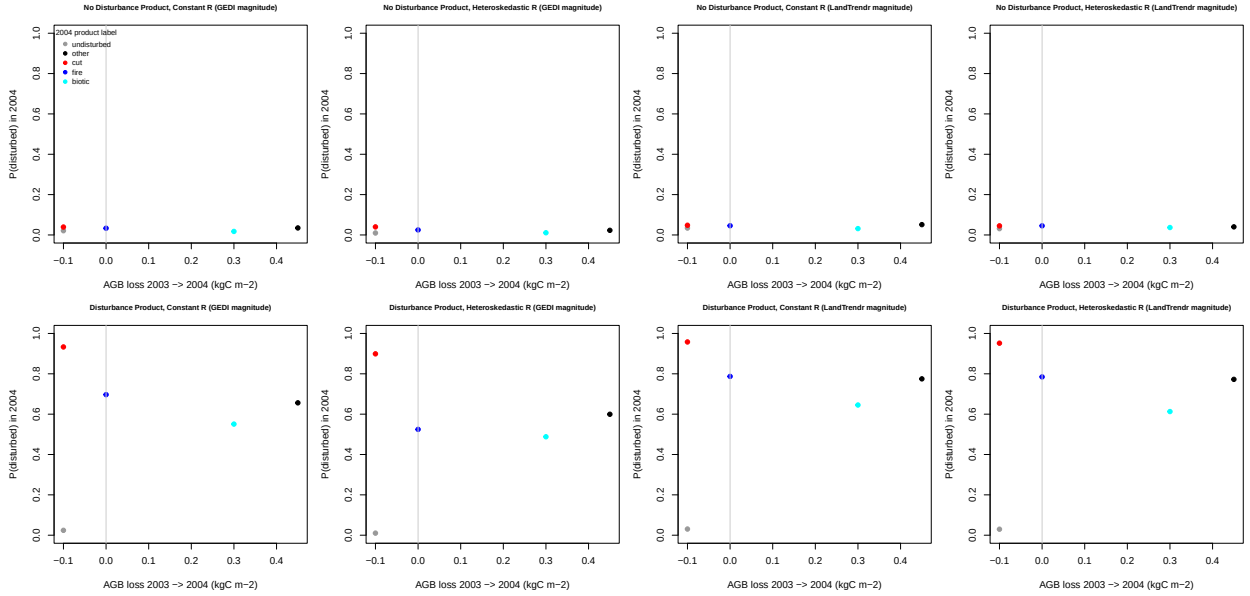


Figure 2: Detection probability in 2004 vs. AGB loss 2003→2004 (test run).

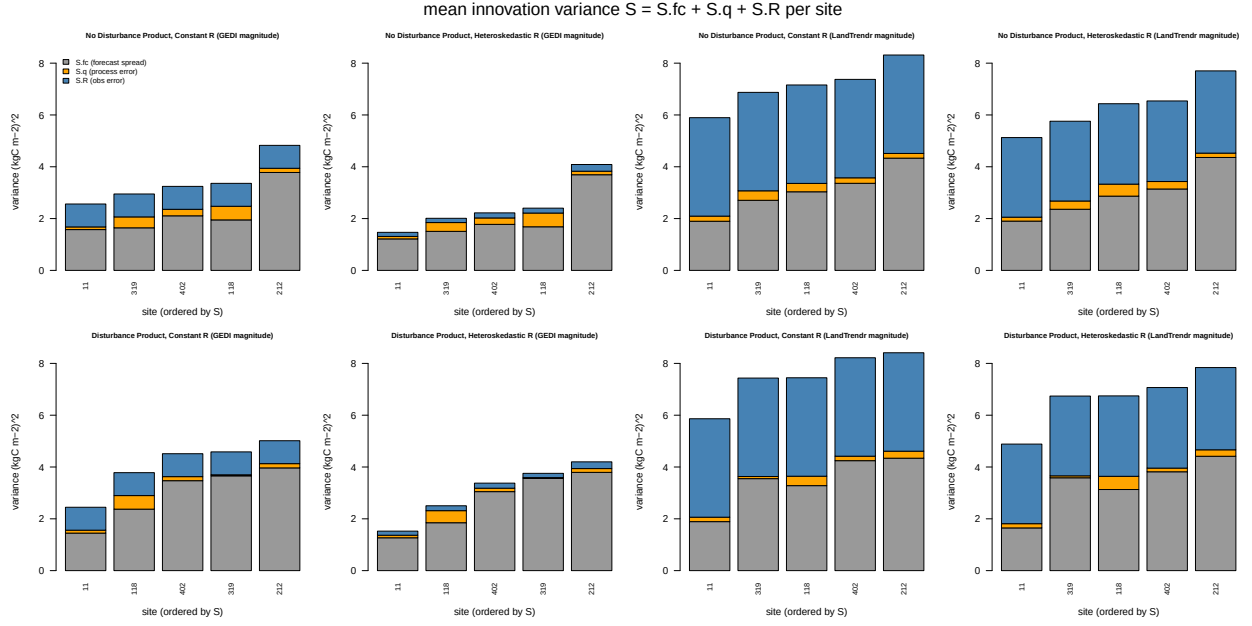


Figure 3: Per-site mean innovation variance split into S^{fc} , S^q , S^R (test run).

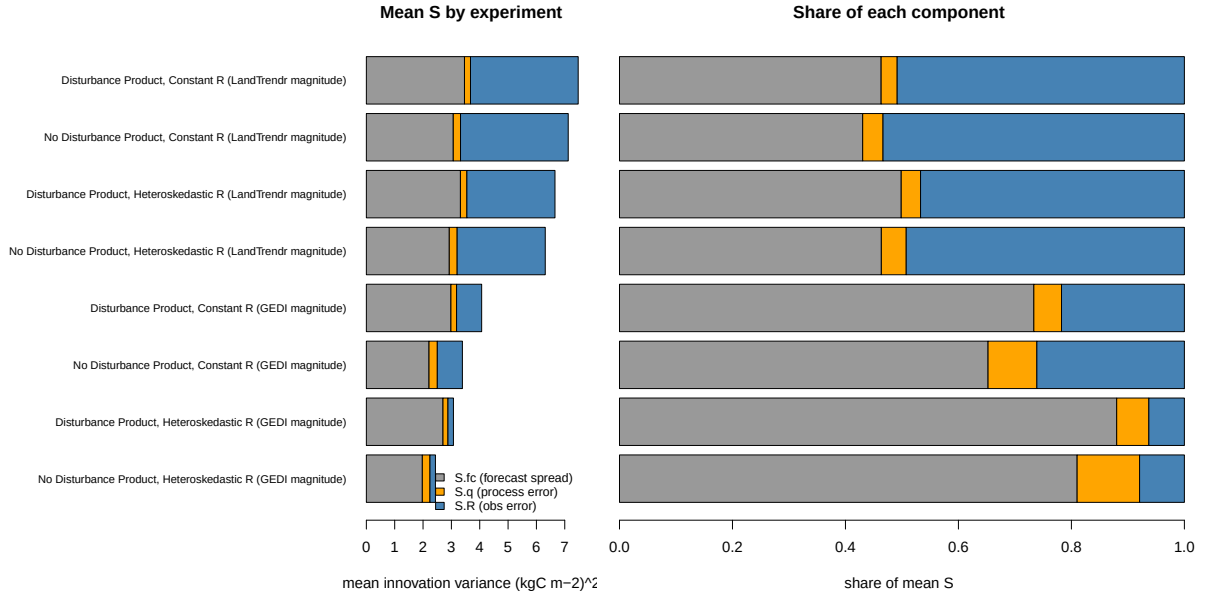


Figure 4: Across-site mean innovation variance per experiment, absolute and share (test run).